

## Group Assignment 03

Provide the answers to the following and then submit your document:

1. In this week's reading assignment, Poppendieck and Cusumano discuss how lean principles were applied by Microsoft before the advent of Agile methods like XP and Scrum. Please describe Microsoft's implementation of lean principles based on the Poppendieck paper.
  - a. Microsoft utilizes what is called "Sync and Stabilize" processes where engineers integrate production and debugging daily. This practice enforces the continuous integration of code which in turn allows for instant feedback resulting in immediate correction. This is similar to Toyota's lean methodology of halting production when error or scrap is detected. Additionally, Microsoft emphasizes reducing bureaucracy and reduced staffing which allows for a streamlined approach to decision-making and more efficient progress throughout the development process. These two practices integrate into the culture developed around the focus on quality and continuous improvement. Daily builds and instant feedback contribute to the method of building quality from the beginning of production to release that Microsoft employs for their project management. Comparatively, Toyota utilizes a similar project overview as they focus on Just in Time (JIT) release and minimizing waste in order to assure quality as Microsoft does. By minimizing inventory (in Microsoft's case production time), both companies' daily builds support incremental development by

regularly allocating resources to the creation of smaller features that build an overall product. Thus, Microsoft's implementation of lean principles, as discussed by Poppendieck and Cusumano, showcases an early and influential example of applying lean thinking to software development, setting a precedent for the Agile methodologies that would later become widespread in the industry.

2. Provide an overview of your project team's current process, covering communication, work distribution, code integration, testing procedures, and client result sharing. Subsequently, delve into the concept of **Value Stream Mapping**, and evaluate your team's current process to identify any inefficiencies or waste. Propose a new process that eliminates these issues.
  - a. The project team, tasked with the creation of an AI-powered lie detection application, operates within a framework grounded in Agile methodologies. Their communication is structured through consistent daily stand-ups and effective use of Jira for issue tracking, ensuring clear alignment and rapid resolution of challenges. Roles are well-defined, with work allocated based on expertise; developers set up the technical backbone, machine learning analysts focus on the AI components, and the UX/UI designer shapes the user experience. Their approach to code integration adheres to Continuous Integration principles, with frequent merges into a central GitHub repository, which promotes a stable and progressive code environment. Testing is presumed to be methodical, potentially adopting a Test-Driven Development strategy, assuring quality and functionality

before features reach production. The team's process reflects a commitment to agility, efficiency, and quality, always with an eye towards meeting client needs and advancing the product's value.

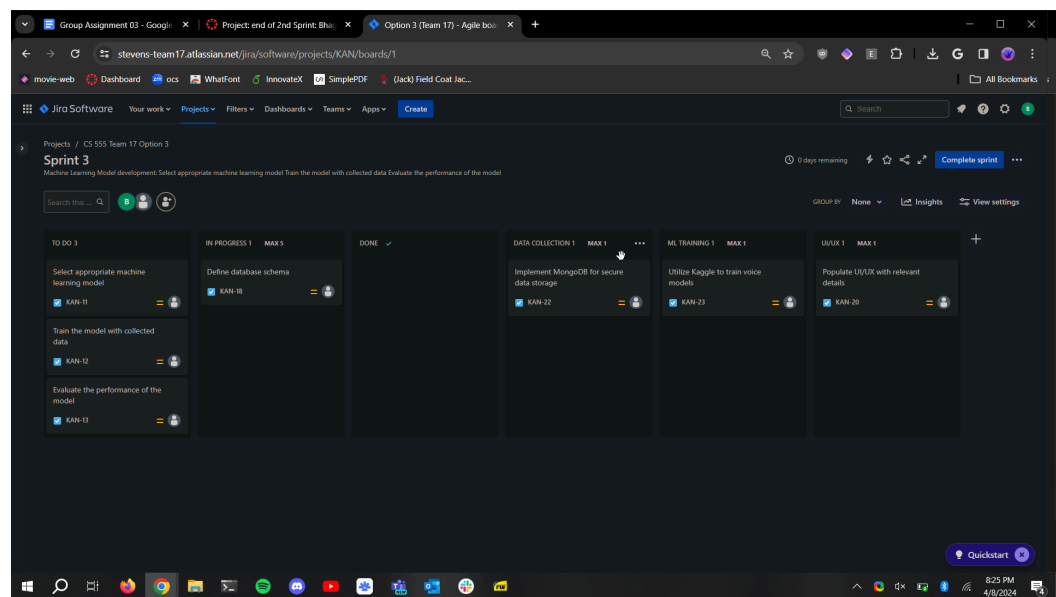
3. Explain the concept of set-based development and share the alternatives your team considered when you first started working on the project if you have.
  - a. Set-based development embodies a strategic approach to innovation and problem-solving by exploring multiple solutions in parallel, thereby fostering informed decision-making. This method allows teams to delay final decisions until critical information is available, ensuring choices are made with a comprehensive understanding of possible outcomes. The hallmark of set-based development is its embrace of flexibility and adaptability, enabling teams to respond dynamically to new insights or changing requirements without being prematurely committed to a singular path. This flexibility significantly reduces the risk of costly late-stage changes or rework, as decisions are grounded in a robust evaluation of diverse alternatives. By incorporating a breadth of possibilities, set-based development enhances the potential for innovative solutions and ensures that final decisions are both strategic and well-informed, aligning closely with lean principles of efficiency and waste reduction.
  - b. At first, the team considered Waterfall development but since the development did not involve any real-time phenomena, we determined there were better iterative methodologies for software development.

Point-based development was also considered but ultimately ignored due to the risk in development that often proves inefficient in the long-term.

The focus of the project was on quality not the speed of production.

4. Describe what a Kanban board is and explore the possibility of combining Kanban with Scrum. Implement a Kanban board for your project in **Jira (Board)**, don't forget to add additional columns and setting column limits. Explain why we should set Work In Progress (WIP) limits. Please provide a snapshot of your Kanban board for submission.
  - a. A Kanban board is a visual management tool that enables you to see the flow of work through various stages of the production process at a glance. It's a cornerstone of the Kanban method, which is a popular lean workflow management system for managing and improving work across human systems. A basic Kanban board is divided into columns, each of which represents a stage in the workflow. Common stages include "To Do," "In Progress," and "Done," but these can be customized to fit any team's process. The work items themselves are represented as cards that move from left to right across the board as work progresses through the workflow stages.
  - b. Combining Kanban with Scrum, known as "Scrumban," harnesses the structured, iterative development framework of Scrum with the flexibility and visual workflow management of Kanban, leading to a synergistic approach that offers numerous benefits. This hybrid model enhances visibility and transparency within teams through the use of Kanban boards,

allowing for real-time monitoring of work progress. It introduces flexibility in task management, permitting adjustments to be made even mid-sprint if necessary, without disrupting the overall workflow. Furthermore, Scrumban facilitates continuous delivery by enabling work to move to completion as soon as it's ready, rather than waiting for sprint boundaries, thus accelerating the delivery of value. Additionally, by identifying and addressing bottlenecks in the workflow, this combined approach leads to improved efficiency and productivity, ultimately fostering a more agile, responsive, and continuously improving development process.



C.

Provide thorough and thoughtful responses to each prompt using your **OWN words**.

Clearly identify each response.